

INTE 423/ BBIT 425 BUSINESS PROCESS REENGINEERING

SUMMARY TEACHING NOTES

- **WHAT IS REENGINEERING:** To reengineer something is to **redesign** it. This can mean an overhaul of something to make it better
- **WHAT IS A BUSINESS PROCESS:** A series of activities with a **start** and an **end** whose aim is to accomplish a business goal
- **WHAT IS BUSINESS PROCESS REENGINEERING:** This is an endeavor to **redesign** and **rebuild** a business process to make it more efficient so as to maximize profit by increasing productivity and reducing the cost of production.
- **WHY REENGINEER BUSINESS PROCESSES:**
 - The aim of reengineering business processes is to align them with the **changing operating business environment** due to technological changes
 - *Currently the business environment is changing towards adoption of computers and other digital devices in the business process*
 - *Computerizing a business without reengineering the business processes will result to an inefficient system*
 - *The purpose of reengineering business processes before computerizing them is to eliminate or obliterate any unnecessary steps*
- **The 3 R's of BPR**
 - *Reengineering*
 - *Redesign*
 - *Rebuild*

The 3 R's of Business Process Reengineering (BPR) are Reengineering, Redesign, and Rebuild. These are key stages in the BPR life cycle.

- **Reengineering** involves identifying and analyzing current processes that need improvement.
- **'Redesign'** is the stage of rethinking and remodeling these processes for optimal efficiency and effectiveness. Lastly,
- **Rebuild** is the implementation stage where the redesigned processes are deployed, transforming how the organization operates to achieve significant performance enhancements.
- **Economies of scale** (mass production at minimal cost of production to maximize profit)
 - **Economies of scale** are cost advantages reaped by companies when production becomes efficient

Overtime there has been efforts to make work easier, reduce cost of production, maximize profit and improve customer experience towards realizing better economies of scale. Some of these efforts focused on: 1) division of labor and specialization, 2) workflow documentation, 3) total quality management (TQM) also called Business process Improvement (bottom up approach in improving work efficiency) 4) Business process reengineering which is about reengineering, redesign and rebuilding of the business process 5) business process modelling/management 6) robotic process automation (automating specific activities in a business process)

- E.g **how does Specialization lead to economies of scale:** As Labor is divided amongst workers, workers are able to focus on a few or even one task. The more they focus on one task, the more efficient they become at this task, which means that less time and less money is involved in producing a good.

- **Specialization vs division of labour:** Specialisation is the process of focusing on a specific task or area of production, while the division of labour is the breakdown of a larger task into smaller, specialized tasks that can be performed by different workers.
- **Workflow** is a series of tasks that must be completed to achieve a specific goal.
- **Workflow automation:** Workflow automation is an approach to making the flow of tasks, documents and information across work-related activities perform independently with minimal human intervention in accordance with defined business rules
- **Business Process Improvement/ total quality management/continuous improvement:** its an incremental approach of improving business processes and all employees have to be involved (*can lead to a heterogeneous IT environment → as a result of implementing sw/hw from different vendors*)
 - ✓ *The cause of heterogenous IT infrastructure especially in midsize companies is due to uncontrolled manner of implementing IT solutions, especially when additional solutions from different vendors are used.*
 - ✓ *The problem with a heterogeneous IT system landscape is that it uses many different communication protocols, data formats, and system standards. This means that the systems and applications have difficulty communicating and exchanging data with each other. The result is isolated information silos, where all enterprise data is isolated from the rest and cannot be connected. The result is labor-intensive, slow, and inefficient business processes that cannot meet the challenges of digital transformation.*
 - ✓ *To solve this problem we enterprise application integration (EAI) technologies e.g APIs*
 - ✓ *Enterprise application integration (EAI) is the use of technologies and services across an enterprise to enable the integration of software applications and hardware systems from different vendors*
- **Business process Improvement (BPI)** is about continuous incremental change and the goal of bottom-up improvement is to include every person at every level of the organization in continuous incremental change. The change can be initiated by the employees
- **Business Process Reengineering (BPR)** is about making fundamental, holistic changes to an operation. As such, it's a *top-down process* in which an executive decision is made to overhaul an entire area of the business. From there, the appropriate team will work to create entirely new workflows to fit into this new foundational mode of operation.
- **Business Process Management (BPM)** is about proper and structured way of monitoring and managing business processes after BPR to keep an eye on the performance and continue improving through BPI. BPM software exist to automate management of business processes. Business process modelling is used to model the business processes and monitor their performance
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- **Robotic process automation:** *this is about automating repetitive activities in a business process using robots.*

WHAT ELSE MOTIVATES BPR

- ***DIGITIZATION*** of business data has resulted to the need to use the digital data in the business setup (***DIGITIZATION is conversion of analogue data to digital data***)
- When digital data is used to add value to customer service then this is called **DIGITALIZATION**
- ***WHAT IS DIGITALIZATION*** is the process of making use of digital content and digital technologies to improve business performance and add value to products and customer services (***IT IS USING DIGITAL CONTENT AND DIGITAL DEVICES FOR BUSINESS ADVANTAGE***)
- *IF a business is able to embrace digital technologies in all its services, then we say it is digitally transformed. This is called **DIGITAL TRANSFORMATION***
- **DIGITAL TRANSFORMATION:** *this is the endeavor of business entity to embrace digital technologies in all its services*
 - *Digital transformation is the integration of digital technology into all areas of a business, fundamentally changing how you operate and deliver value to customers. It's also a cultural change that requires organizations to continually challenge the status quo, experiment, and get comfortable with failure.*
- There are **FOUR types of digital transformation**
 - **Process Transformation (this is where BPR is)**→ the redesign of business processes to adapt to the changing business operating environment due to introduction of digital technologies in the business operations
 - **Business Model Transformation:** this is redefining the go to market strategy using digital technologies to deliver value to evolving customer needs (***value addition to products using technology***)
 - **Business Domain Transformation:** using technology to redefine products i.e. refocusing the business products by innovation new products using technology (***INNOVATIVELY VENTURING INTO NEW BUSINESS FRONTIER USING DIGITAL TECHNOLOGY***)
 - **Cultural transformation:** *changing work culture changes to embrace digital first culture*
- *Overtime business operations have been changing with the changing technologies. This is well demonstrated by the various industrial revolution.*
 - *In the 1st industrial revolution **steam engine technology** resulted to mechanization of production*
 - *In the 2nd industrial revolution **oil and electricity** led introduction of assembly lines thus further improving productivity and reducing cost of production (***automation*** of production was made possible through assembly lines)*
 - *In 3rd industrial revolution invention of computers leading to introduction of **computerization** this made automation of production using computers possible*
 - **Currently we are in the 4th industrial revolution** where AI and other digital technologies such as IOT, BLOCKCHAIN ETC have become vital to business performance (***disruptive technologies***) --- ***error of smart technologies*** (smart manufacturing, smart farming, smart cities, smart house,)
 - *The upcoming digital technologies which are driving businesses today are called **EMERGING TECHNOLOGIES (DISRUPTIVE TECHNOLOGIES)***
 - **DISRUPTIVE TECHNOLOGIES:** *disrupt normal operations and introduce new ways of doing business*

- *SUSTAINING TECHNOLOGIES: this support and improve business as usual operations. These technologies make the cost of production go higher and the products become better but more expensive*
- *When a business does not change according to the operating environment it is likely to be faced out. In this error of technology advancement businesses which do not change on time to adopt to the digital environment are faced out. This is called **DIGITAL DARWINISM***
- **DIGITAL DARWINISM**: *this is a phenomenon where technology and consumers change faster than some businesses are able to adapt.*
- **LEVELS OF CHANGE TOWARDS ACHIEVING ECONOMIES OF SCALE**
Nb: Economies of scale is entails using better approaches to make work more efficient so as to get better output with minimal effort
 - *Division of labor and specialization*
 - *Workflow documentation and automation*
 - *Business process improvement (**total quality management**)*
 - *Bottom-up process meaning it starts from the employees*
 - *Less costly compared to BPR*
 - *Refines and/or automates existing processes*
 - *Low risk*
 - *Incremental*
 - *Takes less time*
 - *Business process reengineering*
 - *It is Top down meaning it starts with top management*
 - *Expensive*
 - *Entails total overhaul of a business process*
 - *High risk*
 - *Time consuming*
 - *Robotic process automation*

The BPR process

The 3 R's of Business Process Reengineering (BPR) are Reengineering, Redesign, and Rebuild. These are key stages in the BPR life cycle.

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The 3 C's of BPR are the forces that drive BPR. They include:

- **Customer:** *sophistication of customers who have become techno privy and are demanding more for less*
- **Competition:** *companies have to keep up with local and global competition in order to remain competitive*
- **Change:** *change in work environment driven by rapid technological changes and digitally informed customers*

Technological Changes has been at the center of all industrial revolutions:

- *Industry 1.0 driving change was steam power leading to steam engines*

- Industry 2.0 driving force was electric power and fuel leading to mechanization of work
- Industry 3.0 driving technology force was the computer leading to computerization of work
- Industry 4.0 driving technology force is smart technologies due data science, AI, IOT and other emerging technologies

STEPS OF BPR

1. **Vision:** define where you want to go
2. **Hypothesizes:** what need to be done to get where the company wishes to be
3. **Process mining:** identify the business processes that need to be reengineered
4. **Design and model:** apply Michael hammer principles to design and model the new business process
5. **Simulate:** use a process mining software to simulate the redesigned business process and assess how it is performing
6. **Go live :** put new process into action

Michael Hammers SEVEN principles of BPR

- 1) Organize around Outcomes, not Tasks :

The job of a person or a department should be re designed by modifying objectives or final outcomes and not by framing their tasks.

- 2) Have those who Use the Output of the Process Perform the Process :

The end users of the output of the processes should perform them themselves. This can be done with the aid of computer based information systems.

- 3) Subsume Information-Processing Work into the Real Work that Produces the Information :

The people or department that generate the raw data should also process the same for use rather than merely sending it to other departments.

- 4) Treat Geographically Dispersed Resources as though they were centralized :

The organisation should use modern information systems to have elasticity in providing services while at the same time keeping the resources to a centralized place.

- 5) Link Parallel Activities Instead of Integrating their Results :

Activities which perform different tasks independently, and are ready to come together, will bring about better integration of the organization's activities.

- 6) Put the Decision Point wherein the Work Is Performer and Build Control into the Process :

The people who are the actual doers should be the actual decision-makers. They should also have sufficient control of their activities.

- 7) Capture Information once and at the Source :

The units should not be engaged individually in database creation. Useful information should be put on network and shared for the benefit of all.

How to reengineer

1. *Document the as is status (how the current system works)*
2. *Document the to be business process (**vision**)*
3. *Document what need to be done to achieve the vision*

4. *Perform process mining:*
5. *Design and model*
6. *Simulate*
7. *Go live*

BPM (business process management)

After reengineering there is need to monitor how the business processes are performing and keep improving them (continuous improvement is what we refer to as TQM or BPI). This does not involve the obliteration stage of process mining.

The business processes are improved as is without redesign.